

SPEC - the Hole Overview page: the map, the photo, and the numbers for the hole

Fable, 2026-09-28, at xhigh (Matt set it: "effort is up"; `get_session self read xhigh`). Revised the same day with his rulings on every number (Section 2), at xhigh (his words: "effort is xhigh again"; `get_session self read xhigh`). Written for Opus to build at xhigh (`.claude/agents/opus-xhigh.md`). Fable reviews the diff against this file, part by part, at xhigh, and signs in `docs/DECISIONS_LOG.md`.

Status: his words, 2026-09-28: *"Make all these edits in the spec and wrap this session up. I will build everything in a new one"*. The session that wrote this built nothing and made no repo writes. Section 13 is for the session that builds.

Revision 2, 2026-09-28, the building session ("Latest build setup"), at xhigh (his words: "effort is up"; `get_session self read xhigh`): the three readings of ruling 16 and the deploy line are answered by him (Section 2, rulings 17 to 20). Two of the three readings were wrong. Changed: Page 1, Section 2, 6.3, 6.6 test 10, Section 8, Section 13.

Page 1 - what you get

What you get. One control on the play screen, MAP, opens a page for the hole you are on. The page shows the aerial photo of the hole with the map drawn over it, and one table of numbers in two columns: **TEE** (from the tee box you are playing) and **YOU** (from where you stand, live GPS). PLAY in the same spot takes you back. Nothing about the round changes when you go there or come back.

On the page	What it reads
Green	front, center, back
Bunkers	to reach it, and to carry it
Water	the creek where a hole crosses it (holes 7, 11, 15, 16): to reach it, and to carry it. Ponds get no number
Creek carry off the tee	always on the page on holes 15 and 16, blue and gold together; on hole 16 the crossing short of the green
Layups	the numbers you type in on the phone, kept on the phone and in the export
Picture	the photo, the hole's shapes, a numbered marker on every bunker and creek crossing, your position

What is decided. Every type of number on the page and how it is computed is your ruling (Section 2). The three readings of your words that were Fable's are answered by you (rulings 17 to 19).

Nothing is built, pushed or installed by this document. The session "Yesterday's round review" had a writer in the working tree when this was filed (the pin sheet, `docs/SPEC_hole-position.md`, builds v33 and v34, from 1692724). On your word this build waits for that session to finish. The install is on your word.

His words that bound it, verbatim, 2026-09-28:

- "There is another session running in this repo so do not touch it with out talking to the other session first. I want to integrate the map in to the app and have a "hole Overview" page I can toggle to. It should have approach distance (front center and back), lay up distances that I will add in later, bunker numbers (carry ideally), distance to the water and carry distance over water."*
- "you are building it that is why I started this wait for that session to finish and effort is up"*
- "remember that book is old. the course map I approved holds above it"*
- "I want you to prompt on each type of number you are putting in the spec and how they should be computed"*
- "add the carry number for the creek off 15 tees and the carry number for the creek on 16 should always be displayed on that hole as well. Make all these edits in the spec and wrap this session up. I will build everything in a new one"*

Standing, from the course map: *"The OSM map is correct I looked it over"* (2026-09-13); *"yes to 10 based off OSM"* (sand from OSM's bunkers); *"yes to 9 I already confirmed that with you. 12 is a single long tee box. Golds towards the center blues at the back"* (2026-09-15).

Provenance tags: `[measured]` read from code or data this session, `[design]` this spec's decision, `[verify]` Opus confirms during the build. Every number carries its n.

Design rules that bind every line (repo CLAUDE.md Section 3): the golfer is the source of truth about where he is, the app only suggests; measured and inferred are never silently mixed; propose and confirm, never detect and fill; nothing is guessed into course data.

0. What this is, and what it is not

Is: six parts, built and committed in this order, the suite green after each:

- **Part A** - the numbers: new pure functions in the course-geometry engine.
- **Part B** - the pictures: 18 photo crops and the frame that places a position on them.
- **Part C** - his course notes: where layups are stored. A storage schema.
- **Part D** - the page, the MAP control, and the page from the home screen.
- **Part E** - hole 9's blue and gold tee boxes go into the map data.
- **Part F** - build id, offline list, the phone app's asset copy, the APK.

Order with the other job: one writer in the working tree at a time (Section 8).

Is not: any change to `js/gps/*` (the GPS precision pipeline is field-validated), to `js/analysis/*` (strokes gained), to `js/round/track-analysis.js`, to `js/data/trackstore.js`, to the round record's keys, `SCHEMA_VERSION` or `migrate()`, to the lock, to the recorder or any Kotlin, to `REVISION`, to any stored round. It does not change `toGreen`, `lieAt` or `nearestHole`. In `js/ui/screen-play.js` it adds the hook in Section 6.2 and nothing else. No network: photos and data ship in the APK. Nothing is taken from the yardage book (Section 12).

1. What exists today [measured]

Piece	Where	State
The map	<code>js/data/geometry/veenker.js</code> (210,888 bytes)	43 tee, 30 fairway, 27 bunker, 26 green, 3 water polygons; 18 hole lines of 2 to 5 points
The water	same	2 ponds (37 m and 66 m across) and loway Creek as one polygon of 1,744 points, 6.9 km by 7.3 km
The engine	<code>js/round/course-geometry.js</code>	<code>courseGeometry</code> , <code>lieAt</code> , <code>toGreen</code> , <code>nearestHole</code>
Green on the play screen	<code>paintGreen</code> , <code>screen-play.js</code> 426-459 at 9e1a548	GREEN 152 - F 143 - B 160 - +/-3 yd, from <code>toGreen</code>
The tee for a tee set	<code>mapTeeBox</code> , <code>screen-play.js</code> 165-173	first box that carries the hole and the set; null on hole 9 and for white and red
The photo	<code>docs/course-map/veenker/basemap.png</code>	USDA NAIP 2025, 4200 x 3311 px, 16,350,678 bytes, 0.27 m per px east-west and 0.37 north-south
Storage	<code>js/data/store.js</code> , <code>schema.js</code>	<code>gt:app</code> and <code>gt:round:<id></code> in <code>localStorage</code> ; the export carries app, rounds, tracks
The phone app's files	<code>android/tracker/app/build.gradle.kts</code> 47-61	copies <code>index.html</code> , the manifest, the icon, <code>css/</code> , <code>js/</code> (not <code>js/dev/</code>)

Measured on the map this session, Fable's scratch prototype (Python, the same plane maths as `js/util/geo.js`), $n = 18$ holes:

Finding	Number
Holes whose line crosses the creek	4: hole 7 (359 to 397 yd down the line), 11 (71 to 99), 15 (279 to 297), 16 (19 to 38 and 407 to 432)
How much the creek crossing moves with your line	hole 7, 300 yd down the hole: reach 59 on the line, 35 from 30 m left, 84 from 30 m right
Ponds beside a hole line	hole 3 (1.5 m from the line), hole 5 (2.8 m and 13.7 m), hole 2 (30 m from the green)
Bunker size, longest dimension	7 to 28 m, $n = 27$
Tee box reach (center to farthest corner)	3.8 to 24.4 m, $n = 25$ boxes the TEE column measures from; the box holes 11 and 18 share is 23.6 m

Finding	Number
Photo crops, 18 holes at 0.30 m per px, the frame rule of Section 4.1	1,432,152 bytes as WebP quality 80; every frame inside the photo, hole 5's after sliding 3 m
The map's labels of which hole a bunker belongs to	wrong: not used by this job (Section 10, follow-up 1)

2. His rulings

All answered 2026-09-28, in the chat that wrote this spec. The answer column is his choice as he made it; where he typed words, they are quoted.

#	Item	His answer	Where it lands
1	Measured from	Both: tee and live GPS	6.3, two columns
2	Page content	Aerial photo + numbers	Section 4, 6.3
3	Layups, how entered	Entered on the phone	Section 5, 6.4
4	Green, from where he stands	On your line: a straight line from him to the center of the green on the map; front is where that line enters the green, back is where it leaves; the same numbers as the play screen's GREEN line	3.7, toGreen
5	Numbers from the tee	Map, from the tee box: a straight line on the map from the center of the tee box for his tee set	3.5, 3.7
6	Bunkers	Nearest / farthest point: reach is the nearest edge, carry is the farthest point, straight line, no aim line assumed	3.7
7	Creek crossings	Your line to the target: a straight line from where he stands to the next point of the hole (the corner, or the green center)	3.6, 3.7
8	Ponds	<i>"ignore"</i> , asked which he meant: No numbers for ponds	3.4
9	The creek beside a hole	No number	3.4
10	Layups, measured from	Both, you pick per number	3.8, 6.4
11	Which bunkers	Greenside to one hole	3.4
12	Hole 9 tees	Yes, add them	Section 7
13	Accuracy	In the column header; numbers are whole yards	6.3
14	The page from the home screen	Yes	6.7
15	The yardage book	<i>"remember that book is old. the course map I approved holds above it"</i>	Section 12
16	Creek carry off the tee	<i>"add the carry number for the creek off 15 tees and the carry number for the creek on 16 should always be displayed on that hole as well."</i>	6.3

Answered 2026-09-28 in the building session's chat, each the option he picked from a prompt that showed the example values:

#	Item	His answer	Where it lands
17	Creek carry line, which tees ("off 15 tees")	Blue and gold together: both every time, whatever tee the round is on. Hole 15 reads CREEK CARRY BLUE 255 - GOLD 223	6.3
18	Creek carry line, where ("always be displayed on that hole")	Hole Overview page only: fixed under the top bar of the page on holes 15 and 16. The play screen is unchanged	6.3
19	Hole 16, which crossing ("the creek on 16")	Short of the green only: the crossing short of the green, measured from the tee. Blue 477, gold 415	6.3
20	This session's job and the deploy	Build Hole Overview + deploy: the push and the phone install of everything on main (v31 to v35) are the building session's, each on his word	Section 8

The first draft's readings of 17 and 19 (the round's tee set only; both of hole 16's crossings) were Fable's and were wrong. They are not built.

3. Part A - the numbers (Fable-owned code, built by Opus from this section)

New pure functions. No DOM, no storage. Positions are { lat, lon, accuracyM? }. Distances in the local tangent plane (enuOffset), never haversine. Yards are Math.round(toYards(m)), as toGreen does.

One ruler (his rulings 4 to 7): every number on the page is the straight-line distance from the origin named at the top of its column to a point on the map. The TEE column is the YOU column computed from the tee box center. The two columns run the same functions.

3.1 js/util/polygon.js, additions

- projectOnPolyline(pt, line) returns { distanceM, sM, leg, side }: the nearest point of an open polyline, its distance from pt, its position along the line from the first vertex, the index of the END vertex of the leg it is on (1-based), and side +1 when pt is left of that leg's direction, -1 right, 0 on it.
- segmentRingCrossings(a, b, ring) returns [{ tM, point }], ascending: where the segment a -> b crosses the ring. Built on rayRingIntersections, cut at the segment's length.
- ringGapM(ringA, ringB): the smallest distance between two rings' boundaries; 0 when a vertex of either is inside the other.
- ringSpanM(ring): the diagonal of the ring's east/north bounding box.

3.2 js/round/course-geometry.js, additions

Constants, all [design], each with its evidence:

Constant	Value	Evidence
COMPACT_SPAN_M	150	tells a pond from the creek: ponds 37 and 66 m, the creek 6.9 km (n = 3 water polygons)
GREENSIDE_M	30	see 3.4
BESIDE_LINE_M	35	see 3.4
CORNER_M	30	hole 3's line has a vertex 18 m after its start
CROSSING_MATCH_M	80	hole 16's two crossings are 369 yd apart; hole 7's moves 25 yd at 30 m off the line

3.3 holePath(geometry, holeNumber)

The hole's line, with its last point replaced by the centroid of the hole's green (the same center toGreen measures to). Null for no geometry or an unknown hole.

3.4 holeFeatures(geometry, holeNumber)

The bunkers and creek crossings that get a number on this hole. Fixed per hole: it does not depend on where he stands, so a marker on the picture never changes its name while he walks. Cached per geometry and hole.

A polygon is **compact** when ringSpanM(ring) <= COMPACT_SPAN_M.

Kind	Rule
Bunker, greenside	ringGapM to the green of any of the 18 holes is <= GREENSIDE_M: numbered on the ONE hole whose green is nearest (tie: the lower hole number), and on no other
Bunker, not greenside	numbered on every hole whose holePath passes within BESIDE_LINE_M of any vertex of its ring
Compact water (a pond)	NOT a feature on any hole: no name, no marker, no number (his ruling 8). It is drawn on the picture. No hole path crosses a pond [measured], n = 18 holes, 2 ponds
Water that is not compact (the creek)	one feature for each place the holePath crosses it: an entry point and an exit point. Where it runs beside a hole without crossing, nothing (his ruling 9)

Why the greenside rule: without it the bunkers at the 13th and 16th greens are numbered on hole 2 at 38 to 52 yd from its tee, and the same at the tees of holes 4, 5 and 12 [measured], n = 7 such pairs.

Each feature:

```
{ name: 'B1' | 'W1' | ..., // B1.. and W1.., by ascending sM; ties by String(id)
  id, kind: 'bunker' | 'water',
  mode: 'beside' | 'cross', // a bunker is 'beside', a creek crossing is 'cross'
  side: 'L' | 'R' | 'C', // of the hole path at the feature; 'C' for a crossing
  sM, sMinM, sMaxM, // down the hole path: its center, nearest and farthest vertex
  at: { lat, lon }, // where its marker is drawn (see below)
```

```
entry: { lat, lon },      // crossings only: where the HOLE PATH enters the water
exit:  { lat, lon } }    // crossings only: where the HOLE PATH leaves it
```

at is the ring's centroid; for a crossing it is midway between entry and exit.

Expected result, all 18 holes: Section 11, Table 1 (25 bunker rows, 5 creek crossings, n = 30).

3.5 teeOrigin(geometry, holeNumber, teeSet)

Case	Returns
One or more tee polygons carry the hole and the set	the centroid of the one whose centroid is farthest from the green's centroid; accuracyM = the distance from that centroid to its farthest vertex, rounded to 0.1; source: 'map'; id
None, and a points entry of kind tee carries the hole and the set	that point; accuracyM: null; source: 'markup'; id
Neither	null

"Farthest from the green" is his hole 10 ruling (veenker_confirmed_corrections.json: *"farther of the two blue boxes (560)"*). Hole 10 blue is the only hole and set with two boxes [measured], n = 1.

3.6 playPath(geometry, holeNumber, origin)

His line of play from origin. With H = holePath and p = projectOnPolyline(origin, H): k = p.leg; when k is not the last vertex and origin is within CORNER_M of H[k], k = k + 1. Returns { path: [origin, H[k], ..., H[last]], originS: p.sM }.

3.7 holeNumbers(geometry, holeNumber, origin)

Returns null for no geometry, unknown hole, or no origin. Otherwise:

```
{ hole, originS,
  green: toGreen(geometry, holeNumber, origin),    // the same object the play screen's line reads
  features: [ { name, kind, mode, side,
                reachM, carryM, reachYd, carryYd,
                inside: boolean,                    // origin is inside the polygon
                behind: boolean,                    // feature.sMaxM < originS
                ownLine: true | false | null } ] }   // crossings only; null for 'beside'
```

Mode	reach	carry
beside (bunker)	0 when inside, else distanceToRing(origin, ring): the nearest edge	the distance from origin to the farthest vertex of the ring
cross (the creek)	the distance from origin to where HIS line of play enters the water	the distance from origin to where it leaves

The crossing on his own line: take playPath(...).path, collect segmentRingCrossings leg by leg; when origin is inside the ring drop the first; pair the rest in order as (entry, exit). The first pair whose midpoint, projected on the hole path, lies between sMinM - CROSSING_MATCH_M and sMaxM + CROSSING_MATCH_M is this feature's: ownLine: true. When no pair matches, reach and carry are measured to the feature's own entry and exit (the hole path's crossing) and ownLine: false; the page marks that row.

Why his own line and not the hole's: the creek crosses at an angle. On hole 7 the same carry is 74, 97 or 116 yd from 30 m left, on the line, and 30 m right [measured], n = 3 positions.

3.8 layupPoint(geometry, holeNumber, layup, tee)

layup.ref	The point
'green'	on the hole path, searching back from the green: the first point whose straight-line distance to the green's centroid is layup.yards
'tee'	on playPath(geometry, holeNumber, tee).path, searching forward from the tee: the first point whose straight-line distance from tee is layup.yards

Returns { lat, lon }, or null when no point of the path is at that distance, or when ref is 'tee' and tee is null. Nothing about the point is stored; it is derived on read from the number he typed.

3.9 Part A tests - test/run.js, new group 'hole overview (numbers)'

One focused test per behaviour, sized like the 'course geometry' group. Positions come from the data (centroids, path points, offsets built with `offsetPoint`). Expected yardages are Fable's prototype, an independent implementation; tolerance 1 yd.

#	Behaviour	Test
1	The tee for a set	hole 10 blue is box 1065741882; hole 16 blue is hole16-back-blue, source: 'markup', accuracyM: null; hole 1 white is null; holes 11 and 18 blue return the same box
2	What is numbered	holeFeatures for all 18 holes gives the names, kinds and sides of Section 11 Table 1
3	A greenside bunker has one hole	bunker 1065746512 is on hole 13 and not hole 2; 1065741609 is on hole 16 and not holes 2 or 4
4	A bunker's numbers, and no pond	hole 3 from the blue tee: B1 278 / 293; holeFeatures has no water on holes 2, 3 and 5
5	Crossing from the tee	hole 7 blue W1 360 / 398; hole 11 blue W1 63 / 90; hole 15 blue W1 244 / 255 and gold 211 / 223; hole 16 blue W1 80 / 99 and W2 457 / 477, gold W1 17 / 35 and W2 394 / 415
6	His own line	hole 7, 300 yd down the path: W1 59 / 97 on it, 35 / 74 from 30 m left, 84 / 116 from 30 m right; ownLine: true all three
7	Behind, and inside	from hole 7's green centroid W1 is behind; from a bunker's centroid reachM === 0 and inside
8	Layup points	hole 7 green 100: a point on the path, 100 yd from the green centroid; tee 250 from the blue tee: 250 yd from it; hole 8 green 400: null; tee with no tee: null
9	The green is one function	holeNumbers(...).green deep-equals toGreen(...) for the same origin
10	No map	every function above returns null or [] for courseGeometry(RADCLIFFE)

4. Part B - the pictures

4.1 Generator: tools/course-geometry/build_hole_images.py

Python 3 with Pillow (12.3.0 is installed [measured]). Reads docs/course-map/veenker/basemap.png, basemap_true_extent.json (the TRUE extent, README trap 2) and the generated js/data/geometry/veenker.js. Writes 18 images and one module. Re-runnable.

The frame of hole n [design]:

Item	Rule
Up	the direction from the hole line's first point to the green's centroid: tee at the bottom, green at the top
What must be inside	the hole line; the green; the hole's tee boxes and tee points; every vertex of a fairway, bunker or compact water polygon that lies within 60 m of the hole line
Margin	25 m on all four sides
Shape	width at least 0.50 of height; widen equally both sides when narrower
The photo's edge	when a corner of the frame would fall outside the photo, the frame slides sideways, 1 m at a time, until all four corners are inside
Scale	0.30 m per px, square
File	img/veenker/hole-NN.webp, WebP quality 80

Expected [measured] on the prototype, n = 18: widths 518 to 1,046 px, heights 779 to 2,092 px, 24,188 to 164,292 bytes each, 1,432,152 bytes total; one frame slides (hole 5, 3 m). The generator prints the table and the total. A frame that cannot be brought inside the photo, or that loses any of "what must be inside" by sliding, is a failure, exit non-zero.

4.2 The module: js/data/geometry/veenker-frames.js, generated

```
export const VEENKER_FRAMES = {
  courseId: 'veenker',
  source: { photo: 'USDA NAIP 2025, Iowa Geographic Map Server (public domain)',
    generator: 'tools/course-geometry/build_hole_images.py',
    inputsSha256: { basemap: '...', extent: '...', geometry: '...' } },
  holes: [ { number: 7, file: 'img/veenker/hole-07.webp',
    widthPx: 1046, heightPx: 2092, mPerPx: 0.3,
```

```

origin: { lat, lon },          // the hole line's first point
up: { east, north },          // unit vector
u0M, v0M,                     // the frame's lower-left corner, metres along up and along right
bytes, sha256,
control: [ { name: 'lineStart', lat, lon, x, y },
            { name: 'greenCentre', lat, lon, x, y } ] ], ... 18 ],
};

```

4.3 In the engine

- `courseFrames(course)`: VEENKER_FRAMES when `course?.geometry === 'veenker'`, else null.
- `framePx(frame, pos)` returns { x, y } in image pixels, y down: $o = \text{enuOffset}(\text{frame.origin}, \text{pos}); u = o.\text{east} * \text{up.east} + o.\text{north} * \text{up.north}; v = o.\text{east} * \text{up.north} - o.\text{north} * \text{up.east}; x = (v - v0M) / mPerPx; y = \text{heightPx} - (u - u0M) / mPerPx$.

4.4 Part B tests - group 'hole overview (pictures)'

#	Behaviour	Test
1	The frame places a position	every control point projects within 1.0 px of its recorded x, y, n = 36
2	Everything numbered is in the picture	for all 18 holes every feature's at, the green centroid and the blue and gold teeOrigin land inside the image
3	The files are what the module says	each of the 18 images loads, and its natural width and height equal widthPx and heightPx

5. Part C - his course notes: where layups live (storage schema, Fable-owned)

Named "course notes" so that nothing here is taken for the yardage book.

5.1 The record

One localStorage key per course: `gt:course:<courseId>`.

```

{ schemaVersion: 1,          // COURSE_NOTES_VERSION; the round and app SCHEMA_VERSION does not move
  courseId: 'veenker',
  layups: [ { id: 'l_<uuid>',
    hole: 7,                  // course hole number
    ref: 'green' | 'tee',     // what the number is measured from
    yards: 100,               // integer, exactly as he typed it
    teeSet: null | 'blue' | ..., // the tee set, when ref is 'tee'; null when 'green'
    label: null | 'short of creek', // his words, 24 characters at most
    createdAt, updatedAt } ], // ISO
  updatedAt }

```

Why its own key and not inside `gt:app` [design]: `loadApp` returns a fresh `newAppState()` when `gt:app` cannot be read, and `boot()` saves it straight away through `setTheme` [measured] (`store.js` 72-74, `app.js` 78 and 386).

Settings come back as defaults and the course model is rebuilt from the rounds; typed layups would be gone. A key that is written only when he edits a layup is not in that path.

Design rule 1 of the data model holds: what is stored is what he typed. No position, no derived distance.

5.2 js/data/schema.js, additions only

`COURSE_NOTES_VERSION = 1; newCourseNotes(courseId); newLayup({ hole, ref, yards, teeSet, label }); migrateCourseNotes(payload)` (v1 is the floor, a no-op with the same shape as `migrate`). `newLayup` returns null for: yards not an integer from 1 to 700; ref not 'green' or 'tee'; ref: 'tee' with no teeSet; hole not an integer from 1 to 18. It trims label and stores an empty one as null. It stores teeSet: null when ref is 'green'.

`SCHEMA_VERSION, newAppState, newRound, newHole, newShot, migrate: untouched`.

5.3 js/data/store.js, additions

Function	Behaviour
<code>courseNotesKey(courseId)</code>	<code>'gt:course:' + courseId</code>

Function	Behaviour
loadCourseNotes(courseId)	the stored notes through migrateCourseNotes; with no key, newCourseNotes(courseId) and NOTHING is written
a damaged value	when the key holds text that does not parse: the text is copied to gt:course:<courseId>:bad:<Date.now()>, the storage-error listeners are told, and empty notes are returned with recoveredFrom naming that key. The damaged text is never deleted
saveCourseNotes(notes)	sets updatedAt, writes through writeRaw; returns false and tells the listeners on a full quota, as every other write does
allCourseNotesIds()	the course ids that have a notes key (not the :bad: copies)

5.4 Export and import

Path	Change
buildExport	adds a sibling courseNotes: { <courseId>: notes } for every notes key. formatVersion stays 1. rounds and app are byte for byte what they were, so a round's hole.pinSheet (docs/SPEC_hole-position.md) rides through untouched
importExport, merge	for each course's notes in the file: a layup whose id is not on the phone is added; one that is already there is left as it is on the phone. Nothing is removed. The report gains layupsAdded
importExport, replace	each course's notes in the file replace the phone's for that course. A course with no notes in the file keeps the phone's
A file from before this build	has no courseNotes key: imports exactly as today
The boot-failure rescue in index.html	already writes every gt: key [measured]; unchanged

5.5 Part C tests - group 'course notes (layups)'

#	Behaviour	Test
1	What is stored is what he typed	newLayup with 100, green: yards === 100, teeSet === null; the five rejections in 5.2 return null
2	Round trip	saveCourseNotes then loadCourseNotes deep-equal; loadCourseNotes with no key writes nothing (key count unchanged)
3	Damaged notes are kept	the key set to {bad: loadCourseNotes returns empty notes with recoveredFrom; that key holds {bad
4	The export carries the notes	buildExport(...).courseNotes.veenker equals the notes; rounds is identical to an export taken before the notes existed
5	Import, merge	a file layup not on the phone is added; one with the same id and different yards leaves the phone's number
6	Import, replace; and an old file	the file's notes replace; a file with no courseNotes leaves the phone's notes alone
7	The round rails did not move	SCHEMA_VERSION === 1; a round saved before and after a layup edit is byte-identical
8	A full quota is loud	saveCourseNotes under a throwing setItem returns false and the listener fires

6. Part D - the page (Opus's UI, to these rules)

6.1 The component: js/ui/hole-overview.js

holeOverview({ course, geometry, frames, teeSet, holes, holeNumber, getFix, onClose }) returns { el, tick, close }. getFix returns the current fix or null (the play screen passes () => ctx.gps.current, which is null when the last fix is older than 4 s [measured]). It reads and writes his course notes through loadCourseNotes and saveCourseNotes. It writes nothing else.

6.2 The hook in js/ui/screen-play.js - the whole of it

Item	Rule
The control	one button, MAP, a fourth control at the right end of the hole navigation row
When it exists	a live round on a course with a map and frames. Not in edit mode: the hook tests editing itself, because docs/SPEC_hole-position.md Section 6.4 gives edit mode a map. Never created for Radcliffe

Item	Rule
What it does	appends the page over the play screen, inside the play screen's own element. The play screen is NOT rebuilt: no <code>ctx.go</code> , no <code>render()</code>
Going back	PLAY on the page, in the same place on the screen as MAP
The heartbeat	the play screen's <code>tick()</code> calls the page's <code>tick()</code> while it is open
The round	opening and closing change nothing in the round and save nothing

Why inside the play screen and not a screen of its own: `ctx.go` rebuilds the play screen, and the capture in progress, the lie still being asked for, and the UNDO banner are in memory [measured] (`screen-play.js` 202-232). A page he goes to on every shot must not cost him those.

6.3 What the page shows

Block	Content
Top bar, fixed	previous hole, HOLE 7 - PAR 5, next hole, PLAY; under it the tee and the card: BLUE - CARD 590
Creek carry off the tee, fixed, holes 15 and 16 only	see "The creek carry off the tee" below
Picture	the hole's photo; over it the outlines of the green, bunkers, water (creek and ponds) and fairways, the hole path, a marker with its name on every feature, T at the tee origin, a tick with its yards at every layup point, and his position with its accuracy ring. A pond has an outline and no marker. Position outside the picture: an arrow at the edge toward it
Table	two number columns, TEE and YOU. Rows: GREEN F, C, B; then each feature down the hole (B1 L, W1), reach / carry; then each layup
Layups	+ LAYUP; each row opens its own edit sheet

Column rules:

State	TEE column	YOU column
Header	BLUE TEE +/-11 yd (the box reach); BLUE TEE (your mark) for a markup point	YOU +/-3 yd (the fix's own accuracy)
No tee on the map for the set	one line: no BLUE tee box on the map	-
No fix	-	one line: no fix; every number a dash
Feature behind him	-	behind
He is inside it	-	IN IT
A crossing not on his own line (ownLine: false)	marked hole line	marked hole line
Layup, ref: 'green'	distance from the tee origin to the layup point	distance from him to the layup point; the row is labelled with what it leaves: 100 out
Layup, ref: 'tee'	the number he typed	distance from him to the layup point; shown only when the page's tee set is the layup's teeSet

The creek carry off the tee (his rulings 16 to 19). On holes 15 and 16 one line sits directly under the top bar of the Hole Overview page and stays there: it does not scroll, it is there with no fix, it is there when he is past the creek, and it is there when the picture fills the screen. It is never replaced by behind. It is not on the play screen (ruling 18).

The line carries blue and gold together, every time, whatever tee set the page is on (ruling 17):

Hole	Crossing	The line
15	W1, the only one	CREEK CARRY BLUE 255 - GOLD 223
16	W2, short of the green (ruling 19)	CREEK CARRY BLUE 477 (your mark) - GOLD 415

Each number is the TEE column's carry for that crossing from that set's tee: `holeNumbers(geometry, hole, teeOrigin(geometry, hole, set))` for set in 'blue', 'gold', the `carryYd` of the named cross feature. Expected values above are Fable's prototype [measured], $n = 4$ tee origins. Hole 16 blue measures from his markup point (`teeOrigin(...).source === 'markup'`), and the line says so: (your mark). A set with no tee on the map reads a dash in its place: BLUE -. Hole 16's crossing in front of the tee (W1, blue 99, gold 35) is NOT on the line. The table is unchanged by rulings 17 to 19: every crossing, W1 and W2 on hole 16, keeps its row and its TEE and YOU numbers for

the page's tee set.

Which holes carry the line and which crossing each reads is a list in the page's code, { 15: 'W1', 16: 'W2' }, with his words beside it; it is not derived. Fit [design]: 16 px or more; when it does not fit 360 px on one line it breaks after the blue number, it does not shrink and does not scroll sideways.

The hole shown [design]: the page opens on the play screen's current hole. The two arrows look at other holes and change only what the page shows, never round.currentHoleIndex.

The picture [design]: a tap on it fills the screen with it; a tap again returns. Filled, each marker carries the YOU numbers (the TEE numbers with no fix). There is no pinch: index.html turns page zoom off on purpose [measured].

When the photo does not load: the outlines are drawn on a plain ground, same markers, same numbers. The map is the data; the photo is a backdrop.

6.4 Layup entry

+ LAYUP opens a sheet: the number (yards, numeric keypad), LEAVES TO THE GREEN or FROM THE TEE, a label (optional), SAVE. A number newLayup rejects is said in one line and nothing is saved. A green layup longer than the hole (layupPoint null) is saved and shown without a point on the picture, its YOU and TEE cells a dash. A row's sheet has the same fields and DELETE; delete asks once (confirmSheet) and then offers RESTORE in the toast, the way UNDO does today.

No single touch adds, changes or removes a layup: every change is a typed number and SAVE, or DELETE and its confirmation.

6.5 Fit and reach, at his page size 360 x 728

Rule	Number
Horizontal scroll	none
YOU green center	40 px, the play screen's size
Every other YOU number	24 px or more
TEE numbers	18 px or more
Tap targets	48 px or more, 8 px or more between them
The top bar	stays while the rest scrolls
The play screen	every existing fold, lie-card and green-line test stays green with the fourth control in the row

Default layout, Opus may change it inside the rules above:

```
+-----+
| < 6 | HOLE 7 - PAR 5 | 8 > | PLAY |
| BLUE - CARD 590 |
+-----+
|      | YOU +/-3 yd |
| photo | GREEN 264 |
| markers | F 250 B 279 |
| T B1 W1 o | W1 59 / 97 |
|      | ----- |
| (tap: fills | BLUE TEE +/-11 yd |
| the screen) | GREEN 547/562/577 |
|      | W1 360 / 398 |
+-----+
| LAYUPS | [+ LAYUP] |
| 100 out | YOU 164 | TEE 462 |
+-----+
```

6.6 Part D tests - group 'hole overview (page)'

Mounted the way 'distance to green (play screen)' mounts (test/run.js 5791-5876 at 9e1a548), with the shipped stylesheet at 360 x 728.

#	Behaviour	Test
1	MAP is always there	present on a fresh hole, after a mark, with a lie pending, on a completed hole; absent on Radcliffe and in edit mode

#	Behaviour	Test
2	One green	with a held fix, the page's YOU green numbers equal the play screen's green line
3	No fix	YOU reads no fix; TEE still has its numbers
4	Going there costs nothing	with a lie pending and an UNDO banner up: open, close; both are still there; currentHoleIndex unchanged; persistRound not called
5	Looking is not moving	the page's next-hole arrow changes the hole shown; currentHoleIndex unchanged
6	Layups through the sheets	add, edit, delete, restore: his course notes change; the round's JSON is byte-identical before and after
7	Fit	the rules of 6.5 on hole 11 (4 features) with 2 layups, on hole 16 (3 features and the creek line), and on hole 1 (none)
8	No photo	the image request fails: outlines and markers are drawn, numbers unchanged
9	One tap back	with the page open, PLAY returns to the play screen on the same hole
10	The creek carry off the tee stays	hole 15 reads blue 255 and gold 223, hole 16 reads blue 477 with (your mark) and gold 415, and neither 99 nor 35 is on hole 16's line; the line is the same on a blue round, a gold round and a white round; the play screen has no such line; it is present with no fix, with a fix on the green, with the table scrolled to its end and with the picture filling the screen; hole 7 and hole 11 have no such line
11	A pond has no number	hole 3 and hole 5: the table has no water row and the picture has no w marker

6.7 The page from the home screen (his ruling 14: Yes)

A COURSE MAP button on the home screen for a course with a map; a screen map in `js/app.js` that mounts the same component with `getFix: () => null`, holes 1 to 18, a tee-set selector starting at `teeByCourse[courseId]`, and PLAY replaced by HOME. GPS is not started, so there is no YOU column. `sw.js` SHELL gains any new module this adds. One test: the screen mounts with no round, shows the TEE column and, on hole 15, the creek line; a layup typed there is in his course notes.

7. Part E - hole 9's tee boxes in the map data (his ruling 12: Yes, add them)

Today no box on hole 9 carries a set [measured], so hole 9 has no TEE column for any tee set.

Step	What
Input	<code>docs/course-map/veenker/veenker_confirmed_corrections.json</code> , hole9 gains <code>blue_tee_osm_id: 1065750754</code> and <code>gold_tee_osm_id: 199289144</code>
Authority	<code>docs/course-map/veenker/README.md</code> , Confirmed facts: <i>"The back box is blue (537 yd along the hole line) and the next box gold (491 yd): his "yes to 9", 2026-09-15"</i> ; the ids are <code>docs/handoff/REPORT_1.1.md</code> line 126
Generator	<code>build_veenker.py</code> rule (2) gives those two boxes their set. The third hole 9 box, 1065727838, stays with no set
Expected diff in <code>veenker.js</code>	the sets of those two polygons and <code>source.inputsSha256.corrections</code> . Anything else: BLOCKED
Test	<code>teeOrigin(G, 9, 'blue').id === 1065750754</code> , gold 199289144; from the blue box the green reads 496 / 513 / 530 and B1 472 / 480, B2 493 / 511; from the gold box 457 / 473 / 490 and B1 432 / 440, B2 454 / 473 (Fable's prototype, tolerance 1 yd); the existing 'course geometry' group stays green
What else it changes	<code>mapTeeBox</code> (shot 1 in end-of-hole entry) starts returning a box on hole 9. That is <code>docs/SPEC_shot-places.md</code> 3.3 working as written. The round-review session was told on 2026-09-28 and agreed: <i>"After your Part E, a map tee on hole 9 gets a position; nothing in D1 depends on it being absent."</i>

8. Part F - build, commit, APK

- **Base:** the hash at which main is clean AND the session "Yesterday's round review" has said it has no writer running. Its revision 4.2 is in (77f89df, v32); its pin sheet job (`docs/SPEC_hole-position.md`, Part A v33 then Part B v34) had a writer in the tree from 1692724 when this was filed. It hands over with one document in `docs/handoff/` and a message that main is clean. Opus does not start before that.

- **The deploy.** Relayed by that session, his words in its chat, 2026-09-28: *"okay you will not deploy this on my phone. When you are done hand it off to the other repo and it will fold this into the map feature build and deploy it."* Confirmed by him in the building session's chat, 2026-09-28 (ruling 20): the push and the phone install of everything on main (v31 to v34 and this build) are the building session's, each on his word. Opus neither pushes nor installs.
- **Files both jobs write:** `js/ui/screen-play.js`, `js/data/schema.js`, `test/run.js`, `test/index.html`, `sw.js`, `js/data/build.js`. That job does not write `js/round/course-geometry.js`, `js/util/polygon.js`, `js/data/store.js`, the map data or the generator [measured] (its Section 10); this job does not write `js/round/round.js`, `js/round/hole-position.js` or `js/analysis/*`.
- **One writer:** opus-xhigh, in the background. Fable makes no repo writes while it runs.
- Commit after each part, suite green each time, staged by name, never `git add -A`. Author `rusty9645@gmail.com`.
- The suite runs at `http://localhost:8123/test/` through `preview_start` name `golf-tracker`. Counts before and after are reported with `n`.
- `BUILD.id` and the `gt-shell-<id>` cache move together, on the LAST commit, to the next id free on main at that moment: v35 as things stood on 2026-09-28 (v33 and v34 are `docs/SPEC_hole-position.md`'s). `REVISION` is not bumped.
- `sw.js` SHELL gains `./js/ui/hole-overview.js` and `./js/data/geometry/veenker-frames.js` (the suite requires every statically imported module). The 18 images are NOT in SHELL: a failed image would fail the whole offline install. On the web build a photo is cached the first time it is seen; the phone app has them in the APK.
- `android/tracker/app/build.gradle.kts`, `copyWebAssets`: one line, `from(File(repoRoot, "img")) { into("img") }`. No other change under `android/`.
- After the last commit: `cd android/tracker && .\gradlew.bat assembleDebug`. Report the APK path, size and `versionName`, and that `img/veenker/` is in it (18 files). Do not install.
- Return `DONE <hash>` with: parts built, the generator tables, the image byte total, suite counts, the APK line, and three screenshots at 360 x 728 (holes 1, 11 and 16, page open, a held fix). Or `BLOCKED <question>` at the first thing this spec does not answer. Never guess a course fact.
- **Must not touch:** everything in Section 0's "Is not"; in `screen-play.js` anything but 6.2.

Cost, stated before the spawn: the two xhigh Opus builds on file ran 20 min / 238k tokens (course map Parts A to C) and 126 min / 610k tokens (native shell), `n` = 1 each. This job has six parts; no build of this shape is on file.

9. Limits, stated

Limit	Number [measured]
The TEE column measures from the CENTER of the tee box	the markers can be 3.8 to 24.4 m from it (<code>n</code> = 27 boxes, hole 9's two included); the column header says so
Blue and gold read the same where they share a box	holes 1, 4, 11, 12, 13, 14, 18
Straight line from the tee is not the card on a dogleg	hole 15 blue: 338 yd to the green center, card 420; the card is shown beside it
A carry over the creek depends on the line	hole 7: 74 to 116 yd across 60 m of width (<code>n</code> = 3)
Ponds get no number (his ruling 8)	holes 2, 3 and 5; drawn on the picture
The creek beside a hole, not across it, gets no number (his ruling 9)	within 45 m of the line on holes 2, 4, 8, 14, 17 at the tee end; it is drawn on the picture
The creek carry off the tee on hole 15 is past the corner	the line of play from the tee box goes to the corner first; the carry is the straight line from the box to where the hole's line leaves the water: blue 255, gold 223
Two mapped bunkers are numbered on no hole	ids 1065730098 and 1065751161, 79 m and 74 m from the nearest hole line
White and red tees	no box on the map carries them: no TEE column
The YOU column is as good as the fix	its own accuracy is in the header
Photo and map are two sources	photo: NAIP 2025; shapes: OSM, which he checked over that photo

10. Follow-ups noticed, not in this job

1. **The map's hole labels on bunkers and fairways are wrong.** `build_veenker.py`'s `centroid()` works in absolute planar coordinates (`rematch.py` line 22) and loses the position: against the true centroid it is off by a median 27.7 m and up to 1,310.6 m; 41 of 57 polygons are off by more than 10 m [measured]. Bunker 1065746149, 11 m from hole 12's line, is labelled hole 2. Labels only; `lieAt` does not depend on them and this job does not read them.
2. `mapTeeBox` (`screen-play.js`) and `teeOrigin` answer the same question. They agree on all 18 holes today (hole 10 blue by list order, $n = 1$). One function should serve both.
3. His markup tips could place blue and gold inside a shared box. His call.
4. The Android back key leaves the app from an open sheet (logged 2026-09-26, his call). The same is true from this page.
5. When the pin-sheet item (D1 of `docs/SPEC_shot-places.md`) is built, the page's green rows can read the pin.

11. Reference tables (Fable's prototype, $n = 18$ holes)

Table 1 - what is numbered on each hole

Hole	Numbered on the picture, down the hole from tee to green
1	none
2	none
3	B1 bunker right
4	B1 bunker left; B2 bunker right; B3 bunker left
5	B1 bunker right
6	B1 bunker left
7	W1 creek crossing
8	B1 bunker left; B2 bunker right
9	B1 bunker right; B2 bunker right
10	B1 bunker left; B2 bunker right; B3 bunker left
11	W1 creek crossing; B1 bunker left; B2 bunker right; B3 bunker right
12	B1 bunker left; B2 bunker right
13	B1 bunker right
14	B1 bunker right
15	W1 creek crossing; B1 bunker left; B2 bunker right
16	W1 creek crossing; W2 creek crossing; B1 bunker left
17	B1 bunker right; B2 bunker right
18	none

Table 2 - the TEE column as the map has it today (yards)

Hole 9 is as it will be after Part E.

Hole	Tee	Box reach m	Green F / C / B	Card	Reach / carry
1	blue, gold	17.1	409 / 421 / 432	435, 419	none
2	blue	11.6	328 / 337 / 346	334	none
2	gold	18.6	275 / 284 / 294	283	none
3	blue	11.1	297 / 304 / 314	333	B1 278 / 293
3	gold	9.0	277 / 284 / 294	289	B1 258 / 273
4	blue, gold	10.0	320 / 329 / 337	349, 340	B1 311 / 325; B2 302 / 312; B3 341 / 350
5	blue	7.5	381 / 394 / 407	402	B1 369 / 386
5	gold	24.4	323 / 336 / 349	350	B1 312 / 329

Hole	Tee	Box reach m	Green F / C / B	Card	Reach / carry
6	blue	14.8	178 / 193 / 210	210	B1 168 / 199
6	gold	9.2	139 / 155 / 172	185	B1 129 / 160
7	blue	9.7	547 / 562 / 577	590	W1 360 / 398
7	gold	9.6	495 / 510 / 525	531	W1 308 / 346
8	blue	7.1	154 / 169 / 183	181	B1 140 / 156; B2 150 / 164
8	gold	12.0	142 / 159 / 173	157	B1 131 / 147; B2 138 / 152
9	blue	5.8	496 / 513 / 530	537	B1 472 / 480; B2 493 / 511
9	gold	15.8	457 / 473 / 490	495	B1 432 / 440; B2 454 / 473
10	blue	3.8	514 / 529 / 542	560	B1 501 / 521; B2 512 / 527; B3 548 / 556
10	gold	15.8	459 / 473 / 486	473	B1 445 / 465; B2 457 / 472; B3 492 / 500
11	blue, gold	23.6	132 / 141 / 150	155, 134	W1 63 / 90; B1 131 / 138; B2 146 / 157; B3 154 / 164
12	blue, gold	13.4	280 / 291 / 300	330, 306	B1 280 / 292; B2 272 / 296
13	blue, gold	18.8	132 / 144 / 156	160, 144	B1 116 / 132
14	blue, gold	13.8	380 / 394 / 408	416, 397	B1 373 / 383
15	blue	6.6	325 / 338 / 351	420	W1 244 / 255; B1 334 / 342; B2 330 / 341
15	gold	10.0	300 / 313 / 326	386	W1 211 / 223; B1 307 / 315; B2 306 / 317
16	blue	his mark	500 / 509 / 519	544	W1 80 / 99; W2 457 / 477; B1 521 / 530
16	gold	13.3	438 / 447 / 457	485	W1 17 / 35; W2 394 / 415; B1 460 / 469
17	blue	8.4	154 / 176 / 196	182	B1 148 / 164; B2 157 / 175
17	gold	16.2	120 / 142 / 162	152	B1 114 / 130; B2 123 / 141
18	blue, gold	23.6	509 / 524 / 540	534, 521	none

12. The yardage book

His ruling 15: *"remember that book is old. the course map I approved holds above it"*. Reason: the book is old and the map is the one he checked and approved. **Nothing in this build is taken from the book:** no number, no aim point, no layup, no tee yardage.

For the record only [measured], 2026-09-28: Fable read the 23 photos in docs/Veenker/ after the first draft. They cover 16 holes; holes 10 and 11 are not among them. The book's bunker and creek numbers sit inside the map's shapes or within 2.2 yd of them (n = 16 numbers, 10 holes). Its green depths are within 4.4 yd of the map's (n = 16 holes, median difference 0.0). This is a check of the map, not a source for the app.

13. For the session that builds this

1. **Confirm with him first:** the three readings under Section 2, and the deploy line in Section 8. DONE 2026-09-28: rulings 17 to 20.
2. **Where this file came from.** It was written in a session scratchpad and never committed. Copies: the project memory folder
C:\Users\Administrator\.claude\projects\C--Temp-gitRepos-golf-tracker\memory\hole-overview\
(this file, its PDF, and Fable's scratch prototype geo.py, engine2.py, frames_spec2.py). Filing it as docs/SPEC_hole-overview.md with its PDF is the building session's first commit, in a clean tree.
3. **The prototype is not the engine.** It is Python, it still numbers ponds (written before his ruling 8), and it exists to give the expected values in 3.9, 6.3, Section 7 and Section 11. The review compares Opus's numbers with these, tolerance 1 yd.
4. **The other session's module.** js/round/hole-position.js (greenFrame, pinFromSheet) lands with v33. This page's green rows stay on toGreen; a "to the pin" row is follow-up 5, not this build.
5. **Cost.** Writing session, 2026-09-28: "Weekly - Fable" read 85 percent at the start and 89 percent before the revision, account-wide with other sessions live, n = 1 read each. Effort xhigh for the draft, medium for the prompts, xhigh for the revision.

14. Revision 3 - stage 1 reviewed, and what stage 2 builds (Fable, 2026-09-28, xhigh)

His pick on the build path, 2026-09-28, asked how to build while the v34 writer was in the tree: "Start now, separate copy". So the build ran in two stages. Stage 1 (Parts A, B, C) was built on branch hole-overview from 6f843ea and merged onto v34 at 2d51b74. Stage 2 is Part E, Part D and Part F, in that order, in the main tree.

14.1 Stage 1 verdict: PASS, one correction owed

Check	Result
Engine, schema, store read against Sections 3 to 5	matches, line by line
Table 2, holes 1 to 8 and 10 to 18, blue and gold	Fable's own run of the engine: every box reach, green F / C / B and reach / carry equals the table, largest difference 0 (n = 34 tee rows)
Pictures on the photo	Fable drew holes 7 and 16 with the map's outlines through framePx: the creek, greens, bunkers and fairways sit on the photo (n = 2, by eye)
Suite on the merged tree, 360 x 728	626 / 626 (Fable, n = 1); 604 at v34 plus 22
Must-not-touch list	nothing on it changed

14.2 Rulings from the review

#	Item	Ruling
R1	5.2 and 5.5 test 1 said "five rejections"; 5.2 listed four	Fable's miscount. The fifth is a label over 24 characters after trimming: newLayup returns null, nothing is cut short. Reason: what is stored is what he typed. The page limits the field to 24 characters, so he cannot type one
R2	A damaged notes key is copied to a new :bad:<ms> key on EVERY load (5.3). The page loads the notes each time it opens	Correction C1, stage 2: loadCourseNotes writes the copy only when no gt:course:<courseId>:bad:* key already holds the same text. Test: two loads of the same damaged text leave one copy
R3	Part E, the input hashes	The generators hash text inputs by their bytes on disk. The main tree holds CRLF copies of two JSON inputs, the repository holds LF, so the recorded hash depended on the checkout. Both generators (build_veenker.py, build_hole_images.py) hash text inputs (.json, .js) with CRLF read as LF; the photo is hashed as it is. Part E's expected diff in veenker.js is then: the sets of boxes 1065750754 and 199289144, inputsSha256.corrections, and inputsSha256.markup_lines moving from faa8e10e... to 6e4f5466... (the committed file's hash). Provenance only; no course fact moves. Anything else: BLOCKED
R4	Imported notes are not validated layup by layup	The page shows only a layup newLayup would accept (same five rules) and skips any other without deleting it
R5	Frames: pixel counts rounded up; hole 18 is 2.1 m taller than the prototype because a fairway inner-ring vertex counts	Accepted: 1,433,580 bytes, 18 files (prototype 1,432,152)
R6	The toast for damaged notes says "Could not save to this device." on a read	Part D words it for a read: Course notes could not be read. A copy was kept.

14.3 Follow-ups logged, not built

1. docs/course-map/veenker/check_answers.py rewrites the corrections file from scratch and would drop hole 9's blue and gold ids after Part E.
2. .gitattributes has no *.webp binary; git detects the images as binary on its own today.
3. The suite leaves 7 gt: keys per run on a clean test origin (n = 1), from groups that were there before this job.